

AlignFace: Human-Aligned Face Similarity Metric with Interpretable Concept Relations

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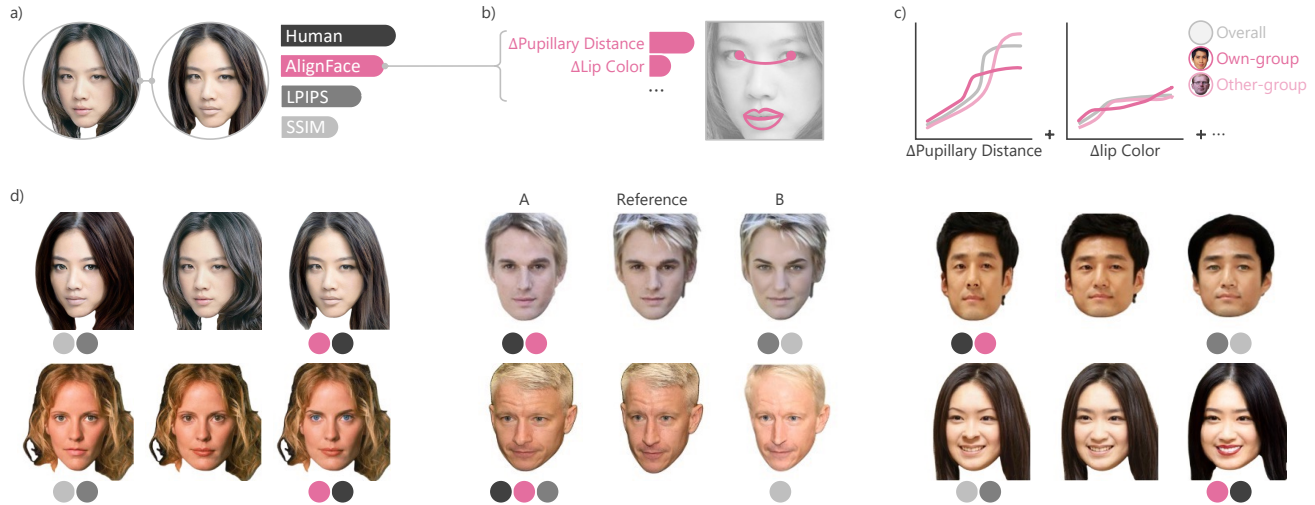


Figure 1: a) AlignFace provides a face-pair similarity score that better matches human perception. b) It decomposes face similarity into attribute-level comparisons, capturing featural (e.g., lip color) and configural (e.g., pupillary distance) cues for fine-grained explanations. c) Decisions are explained via learned nonlinear functions that map attribute similarity to overall face similarity, while capturing group-dependent effects (own- vs. other- group). d) AlignFace outperforms metrics like LPIPS and SSIM in triplet comparisons—whether A or B is more similar to a reference—improving alignment with human perception.

Abstract

Computer vision models for generated facial content, such as face editing and privacy protection, increasingly affect people, requiring similarity metrics that serve as faithful proxies for human perception. While perceptual evaluation has progressed from signal-based heuristics to representation-based metrics, current approaches are limited to behavioral modeling without cognitive alignment. They rely on implicit and spurious relations while assuming a universal observer, failing to account for inherent variations across diverse human populations. This leads to inaccurate evaluative models of stakeholders and misleading guidance for generative model debugging. Rather than treating perception as a black box, we leverage scientific findings from cognitive psychology of human face similarity perception: dependence on facial featural and configural

attributes, nonlinear psychophysical response scaling, and own-group biases. We introduce the FACETS dataset and propose AlignFace, an interpretable, human-aligned, face similarity metric that encodes these cognitive principles through ante-hoc modeling. It employs visual-language modeling (VLM) to encode paired face images and text-based attributes, gated cross-attention (CA) to extract attribute-specific facial difference representations, concept bottleneck modeling (CBM) to constrain reasoning via interpretable face attributes, and neural generalized additive model (GAM) to model their nonlinear influence. Experiments found AlignFace significantly improves alignment with human subpopulation perceptions compared to baseline metrics, including recent domain-free learned perceptual metrics. By bridging learned representations and human cognitive processes, this work enables more transparent and aligned perceptual evaluation metrics for face images.

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CCS Concepts

• Computing methodologies → Computer vision; Image representations; • Human-centered computing → User studies.

Keywords

Face Perception, Face Evaluation, Human-AI Alignment, Similarity, Interpretability, Explainable AI, Vision-Language Model

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1 Introduction

The study of faces in computer vision has given rise to many applications for machine understanding, including face recognition [18, 42, 61, 72], social media retrieval [14, 84], expression recognition for affective computing [16, 71], and descriptive captioning [32, 37]. In these paradigms, faces are treated as data points for classification. However, innovations in generative face synthesis and automated image editing have redefined the prediction tasks.

Recent works target human observers with visual content for human consumption, instead of machine interpretation. These human-centric applications range from face editing [8, 20, 31, 36, 40, 63], including digital makeup [17, 38, 78] and restoration [73, 83] to obfuscation for privacy protection [7, 79, 82]. While standard methods for evaluating model performance involve metrics and models [28, 74], these data-centric approaches are poor proxies for human perception [24], critical for estimating the true effects on people. Thus, a high-performing face verification model could deceptively underestimate privacy risk or identity shifts by identifying two faces as different when a human would still perceive them as similar, or vice versa. Recent efforts to embed human perceptual ratings [85], semantic constraints [50], and conceptual abstraction [49] lack transparency and are not cognitively grounded, risking spurious correlations [58] and being unfaithful to human perception.

Instead, we argue that perception models should adhere to scientific cognitive effects to guide their alignment with humans. For face perception tasks, we focus on the psychology of human face recognition [10]. First, humans process faces based on featural attributes (face parts, e.g., eye color, nose shape) and configural attributes (spatial layout, e.g., pupillary distance, jaw width) [3, 15, 47]. This determines which features or concepts face perception models should encode explicitly or implicitly. Second, humans do not perceive stimuli linearly [66], e.g., perceived brightness varies by the square root of light intensity, and perceived differences depend on absolute values of each stimulus [25]. Hence, we hypothesize that human perception of attribute differences nonlinearly affects their perception of the similarity between two faces. Third, humans have varied backgrounds and abilities, leading to individual or group variances across a population. Notably, own-group bias has been observed, showing that people better recognize faces from their own demographic groups (ethnicity, gender, etc.) [45, 48]. Thus, human alignment should steer model behaviors toward human subgroups rather than assume a universal model for all humans.

We leverage the aforementioned cognitive principles with a data collection experiment and a new human-aligned model for face perception. We collected large-scale human perceptual annotations for both overall face and attribute-specific similarity ratings from two-alternative forced choice (2AFC) comparison tasks [9].

This produced 9.36k triplet ratings of face similarity from 78 participants, and 73.32k triplet ratings of 20 face attributes from 611

participants. Trained on these data and grounded in cognitive principles, we propose *AlignFace*, a human-aligned similarity metric model for face perception. It is ante-hoc interpretable with modules for face attribute predictions and nonlinear attribute-overall perception mapping. Each face attribute is predicted from a contextual fusion of a pair of face images for comparison, and a textual prompt of the attribute (e.g., “eyebrow shape”) [6]. The attributes are combined as a multi-label concept set in a Concept Bottleneck [34] to ensure downstream decisions in terms of these interpretable attributes. These attributes then serve as inputs into a neural Generalized Additive Model (GAM) [12] to capture their nonlinear partial influence on the final overall perception score.

In experiments, we evaluated the human-alignment of AlignFace against heuristic similarity, learned perceptual, and cross-modal baselines. We compared the correlation with human ground-truth ratings for overall and per-attribute similarity, for all humans and for specific ethnic groups, and performed an ablation study on the model modules. Results showed that AlignFace was more correlated to human perception than baselines overall, per-attribute, and per-group. Visual examination of partial dependence from the GAMs also revealed highly nonlinear relations, confirming the hypothesis of nonlinear attribute influence.

Our contributions are:

- 1) **Cognitive characteristics**—featural and configural attributes, nonlinear perceptual scaling, and demographic (White/Asian) own-group bias—identified for human face similarity perception.
- 2) **AlignFace**, a cognitively-grounded, interpretable ante-hoc model for face perception that integrates the cognitive characteristics via Visual-Language encoding, Gated Cross-Attention, Concept Bottleneck and Neural GAM to enforce human alignment.
- 3) **AlignFace2AFC**, an explainable human-aligned face similarity metric that explains perception via the nonlinear contributions of cognitively-grounded attributes.
- 4) **FACETS**, a human-annotated dataset of face triplets with similarity judgments at both overall and attribute levels across 20 cognitively-grounded face attributes.

2 Related Work

We examine why current face models fail to capture true human perception due to neglecting cognitive principles, and how AI interpretability methods can close this gap.

Face Representation Models. Face models have been developed for myriad tasks, such as identity recognition [18, 61, 72], expression analysis [89], landmark localization [77], and attribute prediction [43]. Many depend on face representation model backbones, particularly margin-based architectures like ArcFace [18] and CosFace [72], or triplet-loss based models like FaceNet [61]. However, despite their reliance on human-annotated data, these models remain primarily data-centric, optimizing for categorical ground truths rather than accounting for the subjectivity and nonlinear response scaling inherent in perceptual attributes. This limitation stems from face benchmarks relying on binary annotations that neglect the contrastive nuance and context dependence of human perception [30, 43, 68]. In contrast, our model training collects human perception labels from pairwise 2AFC judgments of overall

face perception and multiple face attributes, and measures viewer demographics to account for perception biases.

Perceptual Similarity Metrics. The advancement of image generation necessitates evaluation metrics faithful to human perception. Conventional metrics like PSNR [28] and SSIM [74] or transformer-based representations like CLIP [56] and DINO [52] fail to capture semantics or fine-grained perceptual nuances, offering poor proxies for human judgment. Learned metrics like LPIPS [85] or DreamSim [23] improve alignment, their black-box nature risks depending on spurious features and correlations [58]. Indeed, DNNs often exploit cues divergent from human perception [24]. While recent work attempts to enforce representational alignment through local-constrained global transform [50] or concept abstraction [49], these methods are data-driven and not cognitively grounded.

Focusing on facial perception, humans prioritize a combination of featural (face parts) and configural (spatial relations) attributes [15, 47], and integrate them additively [5, 70]. Furthermore, their effects are governed by nonlinear psychophysical laws [25, 66] and modulated by observer-dependent variations, such as the own-group bias [45, 48], characterized by superior recognition of one’s own demographic group. Unlike prior metrics that are data-driven and not cognitively grounded, we leverage these cognitive principles to design a human-aligned face perceptual metric that explicitly structures facial similarity through facial attributes, with nonlinear psychophysical weighting and observer-dependent modulation, introducing cognitively grounded inductive biases.

Explainable Vision Models. Ante-hoc interpretability has been shown to improve model performance while supporting human interpretability [59]. Unlike visual explainable AI (XAI) techniques, such as saliency maps [62, 64] that can be spurious [86, 88], such approaches explicitly encode cognitive principles [2, 87] or domain knowledge [41, 46] to improve human alignment. While concept-based explanations [33, 34] can encode human-understandable concepts, they assume these concepts influence behavior linearly. Instead, generalized additive models (GAM) [11, 12] can explain the nonlinear influence of features on predictions. In this work, we developed an ante-hoc interpretable model, leveraging concept bottleneck [34] and GAM [12] together to align perceptual attributes nonlinearly to human perception.

Furthermore, much of explainable AI (XAI) has focused on discrimination tasks, with few methods specifically for similarity tasks. They focus on showing comparative saliency maps with similar regions [91], saliency and similarity [42, 90], and landmarks of positive/negative contributions [21]. Within the facial domain, research similarly focuses on saliency maps [76, 81], and more recently on grouped concepts of face attributes [67] or inferred concepts [54]. While prior methods focus on attribution of latent activations or features, we ground and constrain our approach on cognitive principles, leading to a more theoretically justified model, which we also show to be more accurately aligned with human perception.

3 Face Attributes for Comparative Evaluation with Triplet Similarity (FACETS) Dataset

To develop a human-aligned face similarity metric, we collected human perception ratings of similarity between faces overall and per-attribute for 20 face attributes.

Table 1: Cognitively-grounded featural and configural attributes of human face perception adapted¹ from [3].

Category	Attribute	Scale (0–1)
Featural	Eye shape	Rounded–Straight
	Eye shape	Thin–Thick
	Eye shape	Narrow–Round
	Eye size	Small–Large
	Eye color	Light–Dark
	Nose shape	Pointed–Flat
	Nose size	Small–Large
	Mouth size	Small–Large
	Lip thickness	Thin–Thick
	Lip color	Light–Dark
	Hair color	Light–Dark
	Skin color	Light–Dark
	Skin texture	Smooth–Rough
Configural	Face aspect ratio	Short-wide–Tall-narrow
	Hair length	Bald–Long
	Forehead height	Short–Long
	Pupillary distance	Small–Large
	Cheek shape	Sunken–Puffy
	Jaw width	Narrow–Wide
	Chin shape	Pointed–Square

¹ We omitted ear attributes due to occlusion by hairstyles, and added lip color due to variation in makeup.

3.1 Image Selection and Stimuli Preparation

We constructed our face-pair stimuli using images from two face datasets spanning both lab-controlled and in-the-wild faces: CMU Multi-PIE [26] and CelebA [43]. To better balance demographic groups (gender, ethnicity) from the two datasets, due to the severely limited number of Asian identities (only 40), we subsampled White identities (to 80) to obtain 120 identities (see Appendix Table 3).

For each identity, we selected four source images with varying poses and illumination. This balances intra-identity variability with inter-identity variability across demographic groups. To reduce potential confounding factors, we excluded images containing heavy facial hair or eyeglasses. This initial image set provides seed images that we use to generate edited faces for comparison.

To emulate real-world scenarios, where changes are often subtle to preserve identity (face editing or retouching) or contextual information (privacy protection), we synthesized edited face images using face image generation techniques to produce variations of seed face photos. We employed diffusion-based inpainting [55] for editing featural attributes, landmark-based warping [44] for configural attributes, and GAN-based transfer [51] for hair-related attributes while preserving hairstyle consistency with demographics. Based on detected face landmark points, we estimated attribute values related to length, size, and shape using geometry-based heuristic face anthropometric methods [22]. Color-relevant attributes were estimated from image color histograms from cropped facial components. We ensured that the generated face had the same distribution of attribute values as the original faces (see Appendix Fig. 11). Finally, we further masked the background from the face images to avoid distraction bias during human annotation.

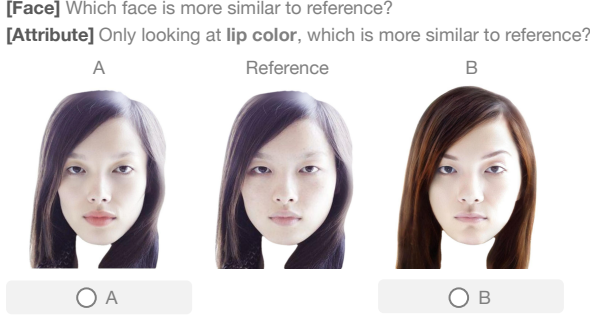


Figure 2: Experiment apparatus of the 2AFC protocol for two user studies on [Face] overall similarity and per-[Attribute] similarity, with different survey questions based on the task.

We constrained the edits to 20 face attributes identified as highly influential in face space [3], encompassing both featural and configural attributes (Table 1). *Featural* attributes describe discrete components or surface qualities that can be identified in isolation, such as the specific color of an eye or the smoothness of skin, regardless of their spatial position on the face. *Configural* attributes describe the spatial geometry and relational distances between features, ranging from internal gaps like pupillary distance to the external silhouette boundaries defined by attributes like cheek and chin shapes, and hair length. Each source image underwent two distinct edits, each editing 3–15 face attributes with randomly sampled magnitudes.

3.2 Psychophysical Similarity Measurements

Having prepared face image data, similar to data collection studies of human similarity perception [23, 85], we elicited human perception ratings via the comparative two-alternative forced choice test (2AFC) [9]. Instead of rating the similarity between two items (A and Target), the participant chooses which of two choices (A or B) is more similar to a Reference [53, 69]. This improves sensitivity to perceptual differences [65] and mitigates response biases [75].

We constructed 480 face triplets (4 random tuples from each of 120 identities) $\langle R, A, B \rangle$, where R denotes the original unedited face, and A and B are two edited versions derived from R . For each triplet, to assess overall similarity, we asked participants to choose “which face is more similar to the reference”; to further assess per-attribute similarity, we asked participants to choose based on one randomly selected attribute out of 20. Fig. 2 shows the experiment apparatus, with different questions for two user studies on overall face similarity and per-attribute similarity.

For both user studies, the experiment procedure is: introduction, consent, screening questions to verify correctness on trivially easy cases, main study with 120 trials each of randomly selected triplets over four sessions interspersed with 30-second breaks, conclusion with demographic questions. Each main study session included one attention check question similar to the screening question. We recruited online participants from Prolific. 78 participants¹ completed the face similarity study in a median time of 23.2 minutes and were compensated £4.00, and 611 participants completed the attribute similarity study in a median time of 30.5 minutes and

¹Although modest, this participant sample yielded 8,880 ratings across 480 triplets, sufficient to fine-tune the pretrained VLM and achieve high agreement (87.0%) with human judgments. (see Fig. 6).

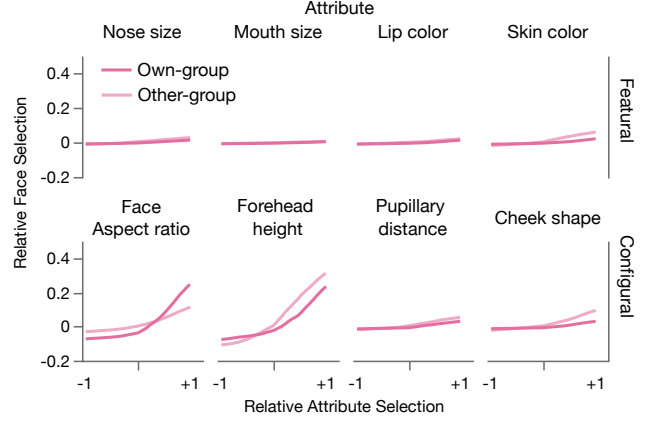


Figure 3: Partial dependence plots of overall relative face selection by the 8 most salient attributes, split by own- and other-group. See Appendix Fig. 12 for full attributes.

were compensated £4.00. Notable for our own-group analysis, we report the ethnicity distribution of our participants²: 377 White, 222 Asian, 55 Black, 11 Hispanic, and 24 Mixed-race. See Appendix Table 4 for participant details. The user studies were approved by our institutional review board.

After collecting the human similarity ratings, to ensure data quality, we excluded 6,510 trials with few ratings (fewer than 3), had ambiguous ratings ($M = 30\text{--}70\%$ selecting either A/B), or were in the same session as a failed attention check question (all 30 responses excluded). Hence, from both user studies, we collected 8,880 triplet ratings of face similarity, and 72,450 triplet ratings of 20 face attributes. We randomly selected 80% of this dataset for model training, and 20% for testing.

3.3 Cognitive Characteristics of Face Perception

To understand which attributes influence face similarity perception more than others, we fit Generalized Additive Models (GAM) with a logit link function. 2AFC selection (A or B) as response and difference in attribute change ($R \rightarrow A - R \rightarrow B$) as factors.

Given the demographic imbalance in our participant sample, we restrict the ethnicity-based own/other-group analysis to White and Asian participants. We fit two GAMs for own-group and other-group demographics to account for subpopulation variance. See Fig. 3 for the results.

Attribute Relevance. Human similarity judgments were unevenly distributed across face attributes; certain features (e.g., face aspect ratio, forehead height) are significantly more influential than others (e.g., mouth size, lip color).

Additive Nonlinear Scaling of Attribute Differences. Perceptual sensitivity to attribute differences is strongly nonlinear. Interestingly, as Relative Attribute Distance Difference increases, i.e., Face B is more different from the Reference than A, Relative Face Selection (toward B) increasingly increases. However, this trend is reversed toward Face A, where the Relative Face Selection effect diminishes with Relative Attribute Distance Difference. This suggests a side-choice bias [80].

²Achieving perfect balance remains challenging due to the inherent demographic skew of Western crowdsourcing platforms [19].

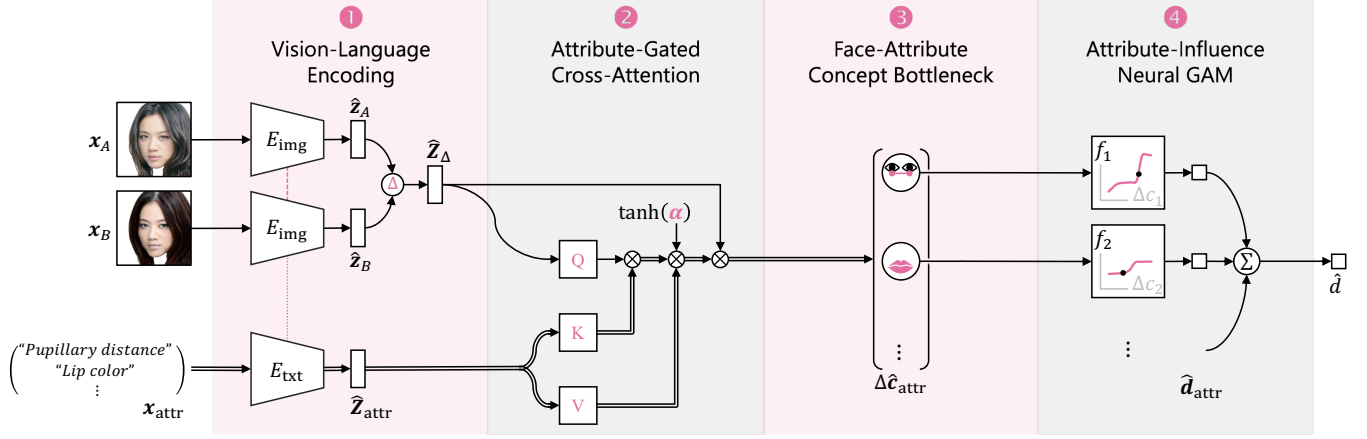


Figure 4: AlignFace architecture. 1) vision-language model (VLM) encoding of face-pair images and text-based attributes into a shared representation, 2) attribute-gated cross-attention (CA) to extract attribute-specific visual features, 3) face-attribute concept bottleneck model (CBM) to constrain reasoning through disentangled face attributes, and 4) Attribute-Influence neural Generalized Additive Model (GAM) to represent each attribute’s nonlinear, additive contribution to the pairwise distance $\hat{d}$.

Demographic Variance (Own-Group Effect). For the White and Asian ethnic groups, different trends were found for own-group and other-group ratings. For example, humans were more sensitive to configural attributes Face aspect ratio and Forehead height, when perceiving faces sharing their own demographics (own-group). Conversely, they were more sensitive to featural attributes Skin color and Cheek shape when perceiving other-group faces. This perceptual divergence confirms that a single, universal similarity metric is insufficient for human-aligned evaluation, motivating the need for demographic-aware steering.

These findings of cognitive effects directly motivate our technical approach, which we describe next.

4 AlignFace for Perceptual Similarity Metric

We propose an interpretable, human-aligned face similarity metric model—AlignFace—for both overall and attribute perceptual comparisons between faces (Fig. 4). Grounded in cognitive principles identified in Section 2 and observed in Section 3.3, AlignFace predicts similarity in a face pair, accounting for multiple face attributes, their nonlinear scaling effects, and demographic-specific variance.

AlignFace leverages 1) vision-language models (VLMs) [39, 56] to map face images and extensible, open-ended attribute-based text prompts into a shared semantic embedding space, 2) gated cross-attention [4] to extract attribute-specific visual features, 3) concept bottleneck model (CBM) [34] to constrain reasoning through disentangled face attributes, and 4) neural Generalized Additive Model (GAM) [12] to resolve overall similarity into an additive composition of nonlinear functions over individual attribute differences. See Appendix Table 8 for justifications of each module in AlignFace compared to standard alternatives.

4.1 Vision-Language Model (VLM) Encoding

While a standard visual encoder extracts embeddings that capture overall facial information, these representations are often semantic-agnostic and thus fail to support attribute-targeted comparisons. We instead leverage a shared semantic representation space from a

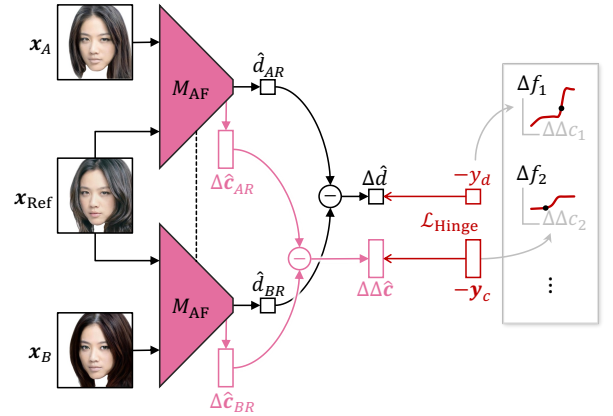


Figure 5: AlignFace2AFC metric using AlignFace (M_{AF}) with a triplet instance $\langle x_A, x_{Ref}, x_B \rangle$ to predict two sets of overall and attribute distances and compute their differences ($\Delta \hat{d}, \Delta \Delta \hat{c}$). Differences are learned via supervised training with binary 2AFC human labels of the overall face y_d or of all attributes y_c . See $y_d - y_c$ GAM-based relations in Fig. 3.

pre-trained VLM to enable face comparison along specific attribute directions (Fig. 4.1). Specifically, we employ a shared vision encoder to extract representations for a face pair as $\langle \hat{z}_A, \hat{z}_B \rangle$. We then apply relational reasoning [60] to model the differences between two face images, yielding $\hat{z}_\Delta^3$. Next, we use the corresponding text encoder to extract descriptions of face attributes (e.g., “Pupillary distance”) into $\hat{z}_{attr}$, which serves as an anchor guiding the comparison between the two faces along a specific attribute dimension.

4.2 Attribute-Gated Cross-Attention (CA)

Having obtained the visual and textual embeddings, we perform multi-modal fusion to extract attribute-specific information for

³Concatenation $A \oplus R$, element-wise difference $A - R$, and Hadamard product $A \odot R$.

face comparison. In particular, we employ Attribute-Gated Cross-Attention (AGCA) [4] on image-difference features as queries (Q) and textual features as keys (K) and values (V), like in latent diffusion models [57]⁴, (Fig. 4.2):

$$\hat{Z}'_{\Delta} = \hat{Z}_{\Delta} + \tanh(\alpha) \cdot \text{CrossAttn}(Q = \hat{Z}_{\Delta}, K = \hat{Z}_{\text{attr}}, V = \hat{Z}_{\text{attr}}), \quad (1)$$

where α is a learnable parameter for gating factor $\tanh(\alpha)$, controlling the attribute-conditioned signal injected into $\hat{Z}_{\Delta}$.

The cross-attention operates analogously to a dictionary lookup, retrieving attribute-relevant visual features from $\hat{Z}_{\Delta}$ conditioned on the attribute context $\hat{Z}_{\text{attr}}$. The residual connection stabilizes optimization by preserving the original representation.

4.3 Face-Attribute Concept Bottleneck Model

For a given face pair $\langle x_A, x_B \rangle$, the attribute-gated CA produces one attribute distance measure per text-based attribute prompt. We denote these as *interpretable* multi-label binary concepts $\Delta\hat{c}_{\text{attr}} \in [-1, +1]$, which also bottlenecks subsequent reasoning (Fig. 4.3), making the architecture a concept bottleneck model (CBM) [34].

However, unlike typical CBMs that input the concepts into an MLP for downstream reasoning, we leverage an interpretable module, described next.

4.4 Attribute-Influence Neural GAM

To interpret the attribute influence while maintaining cognitive-grounding with nonlinear response scaling and additive attribute integration, we represent the relationship between overall distance and attribute distances with a Generalized Additive Model (GAM):

$$F(\Delta\hat{c}) = \sigma(\beta + \sum_{i=1}^N f_i(\Delta\hat{c}^i)), \quad (2)$$

where each f_i is a spline function implemented via a small neural network, capturing both value-dependent attribute contributions and their relative importance across factors. $\sigma(\cdot)$ denotes a sigmoid function⁵, and β is the learned bias term. To ensure perceptual consistency, we further impose a monotonicity constraint, i.e., $\frac{\partial f_i(\Delta\hat{c}^i)}{\partial \Delta\hat{c}^i} \geq 0$, to ensure that higher attribute difference never decreases the overall face distance. We implement GAM as neural layers using Node-GAM [12] (Fig. 4.4).

4.5 AlignFace2AFC Perception Metric Learning

We define AlignFace2AFC, a multi-task triplet metric for similarity perception of a face triplet $\mathbf{x} = \langle x_A, x_{\text{Ref}}, x_B \rangle$ (Fig. 5). For each triplet, we apply AlignFace twice to compare face pairs $\langle x_A, x_{\text{Ref}} \rangle$ and $\langle x_B, x_{\text{Ref}} \rangle$ separately to predict their pairwise distances and calculate the **Relative Distance Difference** for the overall face ($\Delta\hat{d} = \hat{d}_{BR} - \hat{d}_{AR}$) and attributes ($\Delta\Delta\hat{c} = \Delta\hat{c}_{BR} - \Delta\hat{c}_{AR}$).

We collected human ground-truth labels of similar face selection for overall face $y_d \in \{+1, -1\}$ and per-attribute $y_c^i \in \{+1, -1\}$. Each y is +1 if face B x_B is selected as more similar to the reference face x_{Ref} , and -1 if face A x_A is selected instead. Note that the similarity label has the opposite sense of the distance that AlignFace predicts,

⁴This inverted from cross-attention in text-guided VLMs [4], since we require attention in terms of face attributes (V) for downstream concept-bottleneck modeling. The image-diff query (Q) asks which attributes (K) explain the differences between the faces.

⁵An inverse of the logistic link to model the bimodal distribution of human responses.

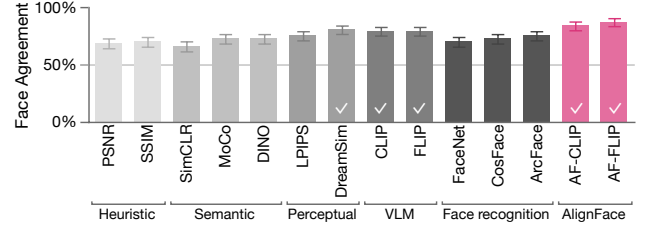


Figure 6: Model-human agreement on overall face perception. The grey line represents random guessing. Error bars show 95% confidence intervals. Baselines without ✓ are significantly lower than AF-FLIP (Dunnett’s test, $\alpha = 0.05$).

so we need to flip it to **Dissimilarity Label** $-y_d$ and $-y_c$. AlignFace is trained by minimizing the squared hinge loss [23] between the dissimilarity labels and relative distance differences, i.e.,

$$\mathcal{L}_c^i(\Delta\Delta\hat{c}_i, y_c^i) = (\max(0, m - \Delta\Delta\hat{c}_i \cdot (-y_c^i)))^2, \quad (3)$$

$$\mathcal{L}_d(\Delta\hat{d}, y_d) = (\max(0, m - \Delta\hat{d} \cdot (-y_d)))^2, \quad (4)$$

where m is a margin (set to $m = 0.05$), to enforce relative ranking via attract-and-repelling.

To manage training instability, we first train the attribute-based difference prediction with per-attribute loss $\mathcal{L}_c^i$, freeze it, and then train the Neural GAM with overall distance loss $\mathcal{L}_d$.

5 Experiments

We evaluated AlignFace to investigate its alignment and attribution faithfulness to human perception, compared to baselines.

5.1 Experimental Settings

5.1.1 Implementation Training Details. We implemented AlignFace in PyTorch, freezing the text encoder and adapting the visual encoder with LoRA [29] ($r = 4$, $\alpha = 8$, dropout = 0.5). The fusion module comprises cross-attention followed by a three-layer MLP (dropout = 0.2), while NodeGAM [12] uses 50 knots. We trained both attribute- and face-level predictions with squared hinge loss (margin = 0.05) using Adam (learning rate = 10^{-3} , weight decay = 10^{-4} , batch size = 16) on NVIDIA RTX 3090 GPUs. AlignFace has 29.42M trainable parameters and required 157.16ms per face pair on a single RTX 3090 GPU.

5.1.2 Evaluation Criteria. To assess alignment with human 2AFC judgments, we evaluate each metric on held-out triplets by comparing its predicted preference with human annotations. Given the relative distance difference $\Delta\hat{d}$ and human labels $y_d \in \{+1, -1\}$ defined earlier, we compute agreement as:

$$a = 1[\Delta\hat{d} \cdot (-y_d) > 0].$$

We evaluate alignment from two perspectives:

- 1) Behavioral Agreement:** We calculate the agreement score between averaged human ratings and binarized model preferences.
- 2) Reasoning Analysis:** To gain deeper insights into the model behavior, we visualize and analyze partial dependence plots (PDPs) to compare the relationship between individual attribute similarities and overall face perception.

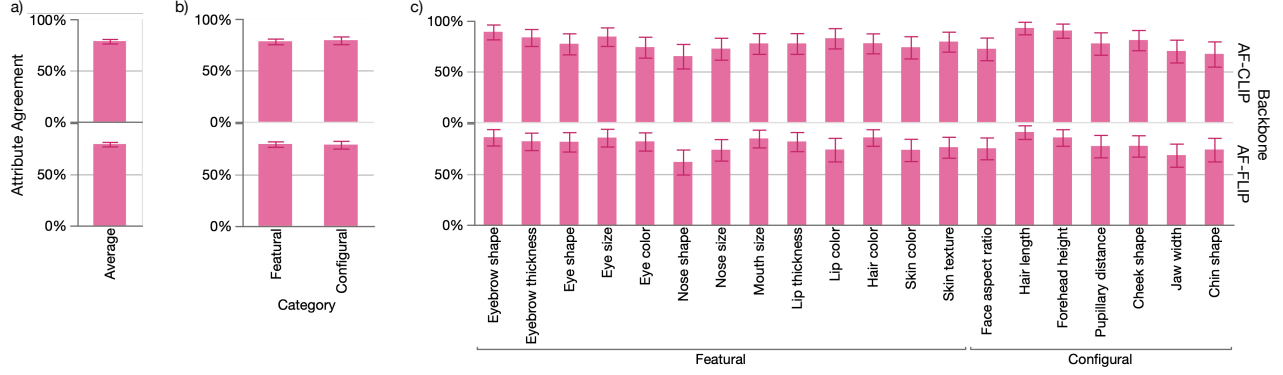


Figure 7: Attribute perception agreement of a) averaged across all attributes, b) grouped into featural and configural categories, and c) at individual-level attributes.

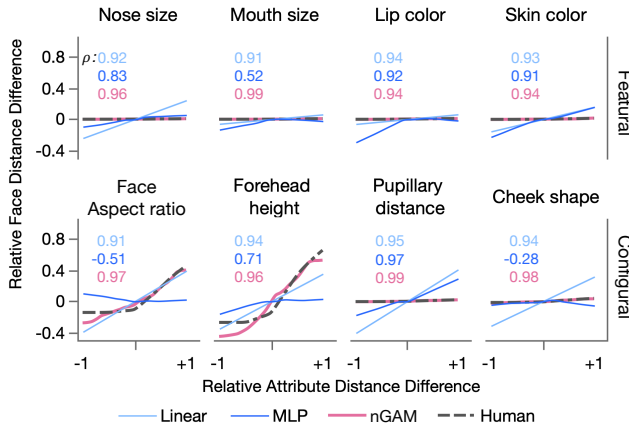


Figure 8: Partial dependence plots (PDPs) of selected attributes, comparing human judgments with Linear, MLP and nGAM. The numbers in graphs are Pearson correlations ρ between each model and humans. High correlations for nGAM (0.94–0.98) indicate strong human alignment of AlignFace.

5.1.3 Baseline Comparators. For overall face similarity, we compare AlignFace against widely used metrics for assessing generative content quality. These include:

- 1) Heuristic metrics: PSNR [28] and SSIM [74].
- 2) Self-supervised semantic visual representation models: SimCLR [13], MoCo [27] and DINOv2 [52].
- 3) Visual-language models: CLIP [56], FLIP [39].
- 4) Perceptual similarity models: LPIPS [85] and DreamSim [23].
- 5) Face-specific models: FaceNet [61], CosFace [72] and ArcFace [18].

We also performed ablation studies to examine the benefits to overall and per-attribute alignment of the modules of AlignFace. For these comparisons, we initialize our fusion module with randomized parameters to provide a rigorous point of reference.

5.2 Alignment Evaluation

Overall Face Perception Alignment. Heuristic approaches (PSNR, SSIM) and self-supervised representation models (SimCLR, MoCo, DINO) exhibit the lowest consistency with human ratings (Fig. 6a). In contrast, perceptual methods outperform both groups, with

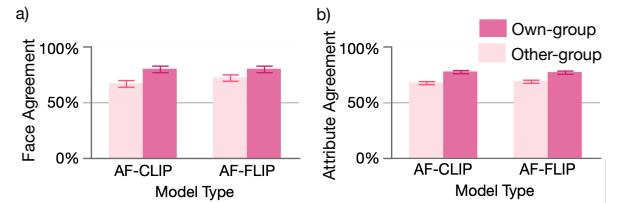


Figure 9: Cross-group perception agreement evaluations on a) face and b) averaged attributes.

DreamSim achieving slightly higher agreement than LPIPS. Surprisingly, approaches based on pretrained contrastive visual-language models (VLMs) achieve relatively strong performance among the baselines. FaceNet, CosFace, and ArcFace baselines performed significantly worse, likely due to their smaller training datasets and lack of VLM pretraining. Overall, AlignFace outperforms all competing methods. We also observed additional gains when using domain-specific encoders (i.e., AlignFace (FLIP) vs. AlignFace (CLIP)), suggesting that FLIP better captures facial semantics.

Face Attributes Alignment. We investigate how well AlignFace aligns with human perception across face attributes⁶. Fig. 7a shows that AlignFace had reasonably high attribute agreement ($M = 0.79$ for FLIP backbone). This was lower than the overall face agreement, perhaps due to attribute diversity and data sparsity. Specifically, it had the highest agreement for Hair length, Eyebrow shape and Forehead height, and the lowest agreement for Nose shape, Jaw width, and Chin shape (Fig. 7c).

Attribute Influence Alignment. Next, we examined whether AlignFace was “right for the right reason” [58] by comparing its attribute influence on overall perception against that of human perception. Specifically, we compared the individual GAM function shapes of AlignFace due to its Attribute-Influence Neural GAM module (Section 4.4) represented in the triple-based dual-pair comparison (Fig. 5) against the human GAM (Section 3.2, Fig. 12). Fig. 8 shows how well AlignFace’s partial dependence plots align with human perception relations, achieving high Pearson correlations ρ . See Appendix Fig. 13 for all attributes.

⁶As a mediation check, we also validated that general-purpose VLMs could accurately encode objectively-verifiable, facial attributes by comparing AlignFace’s predicted attribute sizes and distances against ground-truth heuristic inter-landmark measures [44]. Appendix Table 9 shows high correlations (Median $\rho = 0.947$).

Table 2: Agreement under within- and cross-dataset evaluation for AF-CLIP and AF-FLIP. Consistent performance indicates learned generalized perception across datasets.

Setting	AF-CLIP		AF-FLIP	
	Face	Attr	Face	Attr
Within-dataset	79.9%	77.9%	81.3%	78.3%
Cross-dataset	79.4%	72.9%	81.3%	73.2%

Subpopulation Alignment. Given the own-group bias observed for White and Asian participants in Section 3.2, we next investigated its impact on AlignFace. Fig. 9 shows that AlignFace is in good agreement with human overall perception across the faces of the Own and Other-groups. Notably, agreement was stronger for Own-group perception than Other-group, perhaps because the model was trained on fewer Other-group faces due to imbalanced ethnicities in the face datasets. See Appendix Fig. 15 for results across all attributes. This finding suggests that the proposed approach benefits from tailoring the model to specific subpopulations.

5.3 Cross-Dataset Generalizability

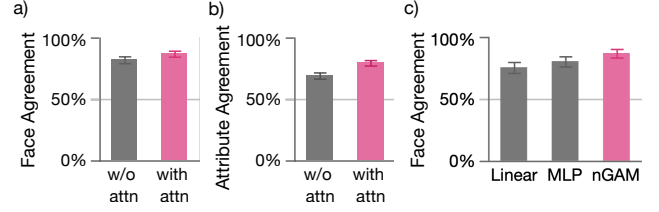
We investigated the impact of dataset domain shifts in face images on AlignFace’s performance. To this end, we retrained two AlignFace models, with CLIP and FLIP backbones, using targets from the CMU Multi-PIE [26] and CelebA [43] datasets separately, and evaluated them under both within-dataset and cross-dataset settings. Table 2 shows that AlignFace achieves consistent agreement scores across both settings. See Appendix Fig. 14 for per-attribute results. This result indicates strong robustness and generalizability across datasets, perhaps due to the VLM’s open-domain knowledge, the use of cognitively-grounded attributes, and the alignment of relationships between attributes and overall perception.

5.4 Ablation Studies

We conducted ablation studies to evaluate the contribution of two key components in AlignFace: 1) Attribute-Gated Cross-Attention for attribute-conditioned representation learning, and 2) Attribute-Influence Neural GAM (nGAM) for aggregating attribute differences into the overall face distance perception. Results are in Fig. 10.

Attribute-Gated Cross-Attention. We replaced the Attribute-Gated Cross-Attention module with a direct fusion variant, where the image-pair and attribute embeddings are concatenated and fed into an MLP. Fig. 10a,b show a clear drop in agreement with human perception at both the overall and attribute levels, with a greater degradation for attribute-level agreement. Even with supervision from human labels, the direct fusion variant fails to capture attribute-specific perceptual cues as effectively. These results highlight the importance of attribute-conditioned visual-semantic interaction for modeling human-aligned similarity.

Attribute-Influence Neural GAM. We further replaced the Neural GAM module with two alternative aggregation heads: a Linear model and a Multi-Layer Perceptron (MLP). The Linear model assumes a simple additive weighting across attributes, while the MLP captures nonlinear interactions in a black-box manner. Fig. 10 shows that the Linear model had the lowest agreement, indicating that simple additive weighting is insufficient to model perceptual similarity. The MLP achieved performance comparable

**Figure 10: Results of ablation studies on Attribute-Gated Cross-Attention: a) overall face model-human agreement, b) average attribute agreement; and on Attribute-Influence Neural GAM: c) overall face agreement.**

to Neural GAM, but lacked interpretability. We also examined the attribute-influence trends using partial dependence plots for the ablated models. Fig. 8 shows the poor fit of the Linear model and MLP. Notably, the complexity of the MLP did not help its agreement compared to Linear. For example, MLP learned spurious trends of decreasing overall perceived difference with increasing relative attribute distance difference for Cheek shape. Hence, Neural GAM had the highest human alignment while being interpretable.

6 Discussion

We have shown the benefits of grounding face similarity modeling with the cognitive principles of featural and configural attributes, nonlinear scaling, and own-group bias. While this work establishes a foundation for human-aligned face similarity, it focuses on controlled perceptual settings with a relatively compact dataset and a specific participant demographic, and excludes ambiguous trials with 30–70% rater agreement⁷. Future work could extend it to broader populations and more diverse generative scenarios, such as privacy-preserving obfuscation, makeup transfer, and restoration.

Beyond faces, AlignFace suggests a general approach for semantic-aware visual similarity. Although we focus on attributes for face perception, the prompt-based VLM can model open-ended domain concepts, such as ABCD criteria for assessing skin-lesion disease progression [1]. Ultimately, our method advances toward human-aligned modeling that mirrors both behavioral responses and underlying cognitive reasoning, which has the potential to inspire other work on human-AI alignment across diverse perceptual tasks.

7 Conclusion

We have presented AlignFace, a human-aligned face similarity metric to model both face and fine-grained attribute perception. Grounded in scientific cognitive principles, it achieves alignment across both predictive behavioral responses and underlying model reasoning. Experimental evaluations demonstrate that AlignFace significantly outperforms existing domain-free metrics in mirroring human judgment. We further provide an in-depth analysis of subpopulation influences, dataset-driven domain shifts, and the architectural impact. Ultimately, this work establishes a more faithful proxy for human face perception, providing a reliable foundation for evaluating generative facial content.

⁷Although common in crowdsourced model training [23, 85], this may omit Point of Subjective Equality (PSE) “hard examples” for fine-grained perceptual differences.

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